3D NACA 0015 wing: secondary experimental comparison

Angle of attack 4 degrees | aspect ratio 6.576 | NASA TP-3151, Fig. 21(a)

Previous solver 0.3.1 | square tip
Earlier diagnostic (revision 8) | square tip
Earlier diagnostic (revision 8) | rounded tip
Current solver (reviewed revision) | square tip
Current solver (reviewed revision) | rounded tip
NASA wind-tunnel data (+/-0.005 rounding only)

A. Lift distribution

Section lift coefficient cl

0.4
0.3
0.2
0.1
0.0

0.0 0.2 0.4 0.6 0.8 1.0

Span position y/s (0 = root, 1 = tip)

B. Pitching-moment distribution

Quarter-chord moment coefficient cm

0.025
0.000
−0.025
−0.050
−0.075
−0.100

0.0 0.2 0.4 0.6 0.8 1.0

Span position y/s (0 = root, 1 = tip)

NASA: Re = 1.5 million, square tip, uncorrected tunnel measurements. Solver: inviscid free air.
Square-tip diagnostic uses span-Q2; rounded tip uses tensor-Q2. Geometry and method both differ.
Rounding intervals are not experimental uncertainty. This is a comparison, not an experimental pass.